CHINMAYA SRI RAMA SESHU PASUPULETI

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EDUCATION

Master of Science in Engineering Science, Data Science **2026 - Present**
University at Buffalo, Buffalo, NY, US, GPA: 3.5/4

Bachelor of Technology in Electronics & Communication Engineering **2017 – 2021**
Amrita University, Bangalore, India.

TECHNICAL SKILLS

- **Programming:** Python, Java, C, R, SQL.
- **Data Engineering & ETL:** AWS Cloud (S3, EC2, SQS), Informatica Power Center, Reltio MDM, dbt, AutoSys, ETL/ELT Pipelines, Data Pipeline Development.
- **Databases & Data Warehousing:** PostgreSQL, MySQL, Oracle, Teradata, Snowflake, Data Modeling, Star Schema
- **Big Data & ML**: Apache Spark, PySpark, scikit-learn, XGBoost, Random Forest, Logistic Regression, ROC-AUC.
- **Visualization & BI:** Tableau, Power BI, Streamlit, Matplotlib, Seaborn, Data Storytelling
- **Tools & DevOps:** Docker, Git, REST APIs, Postman, JSON, Excel

PROFESSIONAL EXPERIENCE

Data Analyst, Cognizant Technology Solutions, Hyderabad, India **2021 – 2025**
Client: AbbVie (American pharmaceutical company)

- Engineered a Reltio Cloud MDM ecosystem for **9M+ HCP records**, **reducing data processing latency by 25%** and achieving **100% data accuracy** through automated AWS and Informatica integrations.
- **Architected** and developed end-to-end **data ingestion workflows** by creating structured **JSON payloads** and integrating **sensitive** data into **Reltio** using **AWS, Informatica PowerCenter, and Java**, while leveraging **Oracle SQL, Teradata, Postman, and AWS services** to optimize and streamline **large-scale healthcare data** operations.
- **Optimized** real-time data workflows via AutoSys, resulting in a **30% reduction in system downtime** and proactive resolution of high-priority technical issues.
- Implemented **30+ List of Values (LOV) configurations** in **Reltio MDM**, reducing **data inconsistency by 15%** across **9M+ HCP records** by designing **LOV frameworks**, enhancing **data structures**, and reprocessing **historical data** to align with **business requirements**.
- Created interactive **Tableau dashboards and analytical reports** by leveraging enterprise healthcare data, enabling executive-level decision-making, performance tracking, and actionable business insights.
(Technologies: **AWS, Informatica, Reltio, SQL, Java, Teradata, Snowflake, Power BI**)

ACADEMIC PROJECTS

Healthcare Admissions Analytics System | End-to-End Data Engineering Project

- Built an **end-to-end healthcare data pipeline** (OLTP → OLAP) using **PostgreSQL, Python, dbt, and Docker** to analyze patient appointment no-show behavior from a real-world dataset.
- Initiated a **normalized transactional schema and analytics-ready star schema**, enabling advanced SQL analysis with CTEs, window functions, and performance-optimized queries.
- Tailored an interactive **Streamlit dashboard** with **KPIs** and **filters** to deliver demographic, temporal, and medical condition insights to non-technical users.

Early Detection of Type 2 Diabetes Using Machine Learning

- Generated ML models (Logistic Regression, Random Forest, XGBoost) on a **253K-record healthcare dataset** to predict Type 2 Diabetes, achieving **81% accuracy with XGBoost**.
- Implemented model evaluation techniques for **imbalanced** data using **ROC-AUC (0.86)** and **PR-AUC**, reducing **false negatives** and improving **prediction reliability**.
- Recognized key health risk factors using feature importance analysis, improving model explainability for real-world healthcare use cases.

Machine Learning Approaches for Early Detection of Cardiovascular Disease

- Built and optimized **machine learning** models (Logistic Regression, Random Forest, XGBoost), achieving **ROC-AUC up to 0.88** using **ensemble techniques**.
- Applied **feature scaling, selection, and cross-validation**, ensuring robust performance on structured clinical data.
- Discovered **key clinical risk factors** through feature importance analysis to support data-driven healthcare decision-making.